Objects of consciousness

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> This is a rewrite condensing concepts from the original paper by Donald D. Hoffman and Chetan Prakash, not original research or a claim of authorship by this blog. The text retains the authors' arguments and perspective, with repeated formulations shortened. Hypotheses, theorem constructions, the conjecture, and proposed physical identifications retain their original status. Equation numbers follow the paper; repeated definitions are consolidated. Corrections and unresolved mathematical points appear in the editorial notes.

About 17,000 → 6,900 prose words · approximately 60% shorter
Abstract + all ten sections · all 21 objections and replies · definitions, constructions, and derivations retained

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Abstract

Standard vision models treat perception as estimating observer-independent physical properties. Evolutionary games and genetic algorithms, however, often eliminate truthful perceptions when these compete with fitness-tuned perceptions: perception guides adaptive behavior rather than estimates preexisting physical truth. Certain quantum interpretations deny definite unobserved dynamical properties; some treat observers as fundamental and wavefunctions as subjective probabilities. These considerations motivate reformulating object perception through a formal "conscious agent" model. We develop interacting-agent dynamics and investigate how objects and spacetime could emerge. A quantum free-particle wavefunction has the same form as harmonic functions characterizing the agents' asymptotic dynamics, permitting particles to be interpreted as agent vibrations, and position, momentum, and energy as properties of their interactions. We sketch possible extensions to relativistic quantum objects and classical macroscopic objects.

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1. Introduction

Humans are predisposed to believe that physical objects retain definite shapes and spatial locations when unperceived. Piaget placed the development of this belief, **object permanence**, between 9 and 18 months; subsequent studies suggested beginnings by 3 months. Einstein's concern about quantum interpretations denying an unobserved Moon illustrates its persistence into adulthood.

Debate has more often concerned *which* properties persist. Democritus treated sweetness, bitterness, heat, cold, and color as conventional, with only atoms and void real. Locke distinguished primary qualities—size, shape, and motion, existing unperceived and resembling our ideas of them—from secondary qualities such as perceived colors and smells, which do not persist unperceived or resemble their underlying causes. By 2014, color remained contested, whereas psychophysics and computational perception rarely questioned the independent existence of shape and position.

Object permanence supports several influential research programs. Physicalist accounts assume that brains and neurons exist unperceived: Edelman locates the emergence of consciousness in brain organization and operation; Crick identifies us with neurons. Standard evolutionary accounts likewise assume independently existing atoms and replicating molecules. James describes evolution as the redistribution of original material, with the same atoms that formed a nebula eventually forming brains.

Computational and Bayesian vision inherit this assumption. Marr treats vision as computing the properties of independently existing surfaces, with evolution progressively improving the representation of objective features. Bayesian theories formulate perception as statistical inference from images about scene properties—including surface shape and reflectance—that exist regardless of observation.

For the authors, the acknowledged failure to produce a plausible theory of consciousness warrants reconsidering such basic assumptions. Yet researchers rarely abandon object permanence. Proposed explanations instead appeal to neural dynamics, ranging from quantum-gravity-induced wavefunction collapses in microtubules to informational properties of re-entrant thalamo-cortical loops, as causing, producing, or constituting consciousness. McGinn expresses the resulting tension: brains are accepted as its causal basis, while how they could generate experience remains unexplained.

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2. Evolution and perception

Is object permanence an insight into objective reality, or a useful habit? If natural selection has shaped our perceptual and cognitive capacities, evolutionary game theory and genetic algorithms can investigate whether—and under what conditions—it favors accurate representations of an observer-independent world (Lieberman et al., 2005; Nowak, 2006; Mitchell, 1998).

Using evolution to question object permanence might seem circular, since conventional evolutionary biology assumes persistent physical objects, including DNA and organisms. But natural selection has an abstract core: variation, selection, and retention. This “universal Darwinism” can be modeled

without assuming object permanence, so the proposed investigation need not presuppose its answer (Dennett, 1995; Blackmore, 1999).

Fodor and Piattelli-Palmarini (2010) challenge natural selection itself, alleging an intensional fallacy in moving from creatures with adaptive traits being selected to their being selected *for* those traits. The paper rejects this criticism, following Futuyma (2010), and regards the evidence for natural selection as overwhelming (Coyne, 2009; Dawkins, 2009).

The standard expectation in vision science is that selection favors **veridical perception**: accurate representation of objective properties that exist when unperceived (Palmer, 1999). The argument is that more accurate perception gave ancestors a competitive advantage, enabling genes for greater accuracy to spread through successive generations; consequently, our perceptions should normally be accurate. This expectation also supports Bayesian perceptual models that estimate pre-existing properties such as shape, position, motion, and reflectance (Knill and Richards, 1996; Geisler and Diehl, 2003).

The reported Monte Carlo simulations of evolutionary games and genetic algorithms instead generally favor strategies tuned to fitness over strategies reporting objective properties (Mark et al., 2010; Hoffman et al., 2013; Marion, 2013; Mark, 2013). Five principles explain this result:

1. **Information costs energy.** Acquiring, processing, and retaining environmental information consumes energy that organisms must obtain. Other things equal, selection favors cheaper perception. Omitting truth that contributes little to fitness can reduce these costs.

2. **Information costs time.** Acquiring and processing more information generally takes longer. Where survival depends on responding before a predator or prey does, speed matters. Other things equal, selection favors faster perception, again making fitness-irrelevant truth expendable.

3. **Fitness need not increase with a world quantity.** Niche adaptation and homeostasis commonly produce nonmonotonic fitness functions: too little or too much salt, water, humidity, or altitude can be harmful, while intermediate levels are advantageous. A perceptual organization tracking fitness therefore need not preserve the ordering of the underlying quantity. In such cases, truth and fitness can diverge, and tracking truth can impair fitness.

4. **Estimating a world state can discard information needed for decisions.** Under perceptual noise and uncertainty, a strategy that first estimates a true state and then acts on that state's utility throws away utility-relevant information. The reported models generally eliminate such strategies when they compete with strategies that use all available utility information without first reducing it to a state estimate (Marion, 2013).

5. **Complexity makes perceptual systems harder to evolve.** Genetic-algorithm simulations show combinatorial growth in the search required to evolve more complex perceptual systems. This difficulty itself creates pressure toward simpler systems, independently of their operating costs.

Selection thus favors perceptual systems whose speed, economy, and fitness sensitivity support survival and reproduction. Genes underlying successful reproduction become more prevalent, whether or not their perceptual systems report objective truth. The intended conclusion is generic, not exceptionless: the paper explicitly allows selection to favor truthful perception under special fitness functions (§9, objection 13).

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3. The interface theory of perception

The evolutionary results challenge the assumption that perceptual usefulness requires accurate representation. How can perception guide successful behavior without revealing objective reality?

A computer desktop provides a model. A blue rectangular file icon lets us open, edit, delete, or copy a file, although the file is neither blue nor rectangular and its bits may occupy dispersed memory locations. Asking whether the icon usefully relates to the file makes sense; treating its appearance as a resemblance to the file confuses their categories. The interface facilitates effective action by hiding circuits, voltages, magnetic fields, and other complexities that would impede the user's task.

It also hides causal structure. Dragging a file icon onto a drive icon appears to cause copying. On this account, the icon's movement itself causes nothing: it guides the user's mouse operation, which triggers an unseen causal sequence. The useful appearance of causation need not disclose the underlying mechanism.

The interface theory applies this distinction to perception. Natural selection fashions a species-specific interface for survival and reproduction: space and time form its desktop, and three-dimensional objects are its icons. Their appearance need not reveal objective structure or causation. A billiard ball apparently causing another to move is, on this hypothesis, another useful interface fiction.

This does not license ignoring perceived dangers. Grabbing a cobra or stepping over a cliff can be disastrous, just as deleting an icon can destroy valuable work. Adaptive behavior requires taking perceptions seriously; it does not require interpreting them literally. Johnson's attempt to refute Berkeley by kicking a stone makes precisely this mistaken inference from consequential interaction to literal representation.

A stronger objection concerns information: if avoiding a perceived cobra protects us, our representation must register something about objective consequences. Fitness-preserving behavior depends on mutual information, in the information-theoretic sense, between predicted and actual utility. Without such information, or a mechanism for increasing it, fitness remains low. We accept this point: perceptual icons convey genuine information about fitness, which is an aspect of objective reality. Gibson's emphasis on affordances—the world's significance for an organism's actions and fitness—is therefore compatible with our approach, although his direct realism and rejection of perceptual information processing are not.

The distinction between truth and fitness requires precision. Let W denote objective world states. For a **fixed organism, its current state, and an action**, a fitness function

$$f: W \to [0, 1]$$

assigns each world state w a fitness value $f(w)$. Raw meat might have high fitness value for a hungry cheetah considering eating, but low value for a hungry cow considering the same action. Fitness is consequently organism-, state-, and action-dependent, rather than a universal measure of the world's properties.

Let m be the true probability measure on W. A fitness-weighted probability measure is

$$m_f(A) = \frac{\int_A f(w)\, dm(w)}{\int_W f(w)\, dm(w)}, \qquad \int_W f\, dm > 0,$$

for measurable events A, so that $m_f(W) = 1$. Given a perceptual mapping, the proposed informational contrast is between maximizing information about world states under m and maximizing it under m_f. These objectives need not coincide. Depending on f, a perceptual system informative about world structure may convey little or no information about fitness, and conversely. This is the intended sense in which perception is tuned to fitness rather than truth: information about adaptive significance does not entail a faithful depiction of objective structure (Cover and Thomas, 2006).

Another objection treats the interface as familiar physics: apparently solid objects are mostly empty space, so their true nature lies in atoms and subatomic particles. The interface proposal goes further. Space and time themselves belong to the interface, so everything located within them—including atoms, leptons, and quarks—also counts as an icon. Object permanence is denied at microscopic as well as macroscopic scales. Replacing familiar objects with smaller physical objects does not escape the interface.

This radical proposal can be situated within a history of correcting the assumption that human appearances reveal reality. Flat-earth appearances gave way to a spherical earth; the apparent stillness of the earth and motion of the heavens gave way to the work of Copernicus and Kepler, despite resistance exemplified by Galileo's forced recantation. Those revisions corrected particular appearances. The proposed next step revises the entire perceptual framework of objects in space-time, treating it as a species-specific mode of perception rather than a disclosure of objective reality.

If these arguments are sound, Bayesian models of object perception require fundamental reconstruction, with consequences beyond perceptual modeling. In particular, a theory denying that objects and space-time exist unperceived or possess causal powers cannot then invoke neurons as the causes of behavior or conscious experience: neurons are themselves such objects. This implication may be received as a reductio of the theory. We instead propose it as a different direction for investigating a mind–body problem that still lacks a plausible explanatory theory.

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4. Definition of conscious agents

Taking the preceding reasoning as our starting point, we explore a mathematically precise theory of conscious subjects as the foundation for a theory of objects. Wilczek's challenge of representing a conscious observer within quantum mechanics indicates the difficulty and scope of this project.

Our method follows Turing's approach to computation. A Turing machine has finite sets of states, symbols, input symbols, and transition rules, together with a blank symbol, a start state, and halt states. The Church–Turing thesis identifies algorithmic computability with computability by such a machine. It is not a provable theorem; an agreed computable procedure that no Turing machine could implement would be a counterexample. None was known to the authors, who describe the thesis as secure, even definitional.

We likewise propose a simple formal object, the *conscious agent*, and a *conscious-agent thesis*: every property of consciousness can be represented by a property of an agent or interacting agents. The aim is a complete theory developed from a small set of definitions and assumptions through theorems and proofs, or simulations where complexity prevents proof. Consciousness is treated on its own terms, rather than derived from a prior physical world. The proposed formalism's scope remains a thesis, formally stated in §6.

An apparent objection is that most mental processing occurs without conscious awareness—the conventional estimate cited here exceeds 90%. We accept the experimental finding that people are unaware of most processing underlying their actions and perceptions. Under the interface theory, however, perception evolved to guide adaptive behavior quickly and cheaply; its finite

representational capacity requires omissions and simplifications. Awareness of all mental processing need no more have been adaptive than awareness of kidney filtration. Inaccessibility to one subject does not establish absence of consciousness: my friend's mind is not directly conscious to me, yet my friend may be conscious. The same distinction applies to processes inaccessible within my own mind. Limits of access should not be mistaken for properties of reality.

This proposal must eventually account for matter, fundamental forces, the Big Bang, the origin and structure of space-time, physical laws, natural selection, and neural correlates of consciousness. We defer these challenges while developing the formalism.

Motivation and channels

The term *conscious agent* receives its content from the mathematical definition below. The following intuitions motivate it but can be discarded without changing that definition.

Three processes form a cycle:

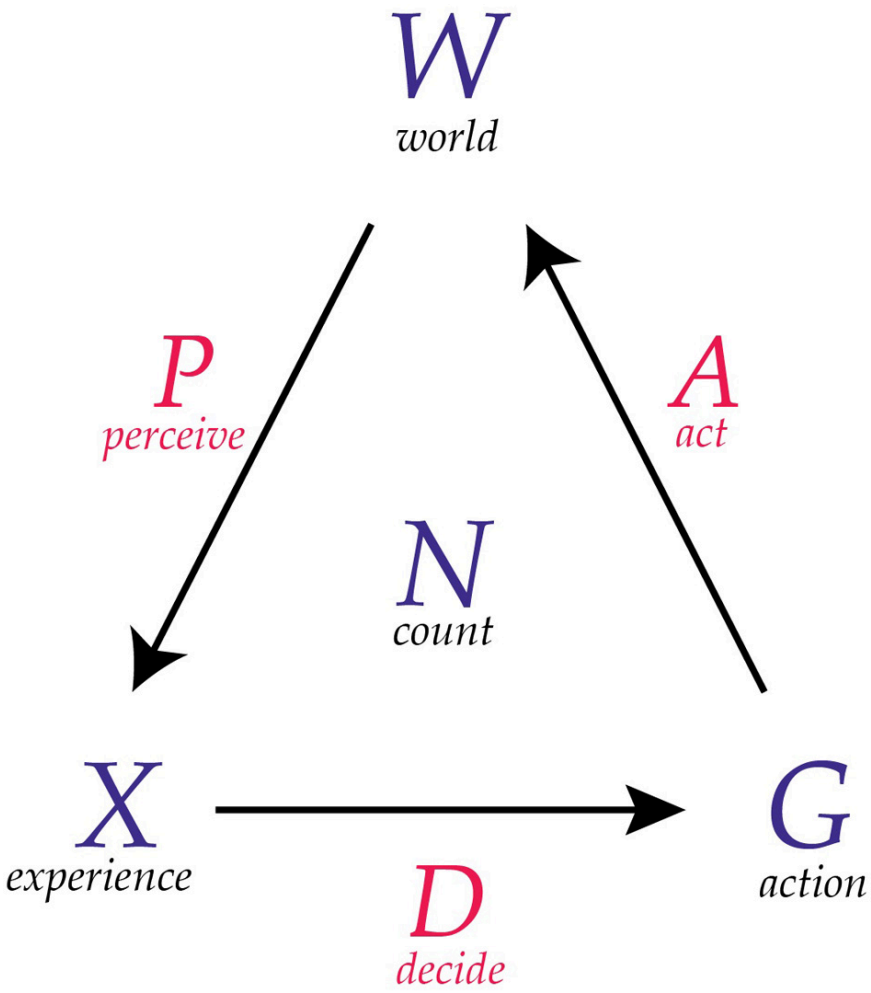
$$W \xrightarrow{P} X \xrightarrow{D} G \xrightarrow{A} W.$$

Perception produces conscious experiences X through interaction with the world W. **Decision** selects actions G on the basis of experience. **Action** affects the world according to those decisions. These arrows will be probabilistic channels, not necessarily deterministic functions.

We seek minimal restrictions. Nearby visual space appears approximately Euclidean, but this does not warrant requiring every experience space to be Euclidean. Likewise, some decisions may maximize expected utility, minimize expected risk, or construct explicit self-models, without making these universal requirements. What the model requires is a way to assign probabilities to experiences, decisions, and effects on the world, conditional on their respective inputs.

A *Markovian kernel* supplies this structure: each input indexes a probability measure on possible outputs. Thus different world states w_1, w_2 can assign different distributions to experience. For finite input and output spaces, the kernel is a matrix of nonnegative entries whose rows each sum to one. Equivalently, it is a discrete information channel with transition probabilities $p(y \mid x)$: the probability of output y given input x.

In this interpretation, perception transmits a message from world to agent; decision transmits one within the agent; action transmits one from agent to world. For discrete interactions, channel uses can be counted. Assuming all three channels operate in lockstep, one integer counter N records the number of messages passed on each channel.



Original Figure 1. One conscious agent: perception P, decision D, action A, experience space X, action space G, and counter N.

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Formal definition

A *measurable space* $(X, \mathcal{X})$ consists of a set X and a collection of subsets called events. This collection is a *σ-algebra* when it satisfies:

1. $X \in \mathcal{X}$.
2. If $E \in \mathcal{X}$, then $X \backslash E \in \mathcal{X}$.
3. If $E_1, E_2, \ldots \in \mathcal{X}$, then $\bigcup_{n=1}^{\infty} E_n \in \mathcal{X}$.

A probability measure assigns probabilities to these events. The formalism starts with measurable spaces; it does not require a single, separately specified probability distribution on each entire space.

More precisely, a Markovian kernel from $(S, \mathcal{S})$ to $(T, \mathcal{T})$ is a map

$$K\colon S\times\mathcal{T}\to[0,1]$$

such that $K(s,\cdot)$ is a probability measure for every $s\in S$, and $K(\cdot,E)$ is measurable for every $E\in\mathcal{T}$.

Definition 1. Given a world represented, for now, by a measurable space $(W,\mathcal{W})$, a conscious agent is the six-tuple

$$C=((X,\mathcal{X}),(G,\mathcal{G}),P,D,A,N), \tag{1}$$

where $(X,\mathcal{X})$ and $(G,\mathcal{G})$ are measurable spaces,

$$\begin{aligned} P &: W\times\mathcal{X}\to[0,1],\\ D &: X\times\mathcal{G}\to[0,1],\\ A &: G\times\mathcal{W}\to[0,1] \end{aligned}$$

are Markovian kernels, and N is an integer. With the σ-algebras understood, we abbreviate this as

$$C=(X,G,P,D,A,N). \tag{2}$$

Each channel P,D,A has a *channel capacity*: the highest information rate, measured in bits per channel use, at which messages can be transmitted with arbitrarily small error probability.

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5. Conscious realism

The world W is needed to define an agent's perception and action kernels. Identifying it with physical space-time would make the formalism dualistic: X and G concern consciousness, while W concerns a physical world. We instead propose a monism in which consciousness is fundamental.

Hypothesis 1 — Conscious realism. The world W consists entirely of conscious agents.

This is a hypothesis that may be wrong. Its implications must be assessed against existing data and theories and developed into novel, testable predictions.

Two conscious agents

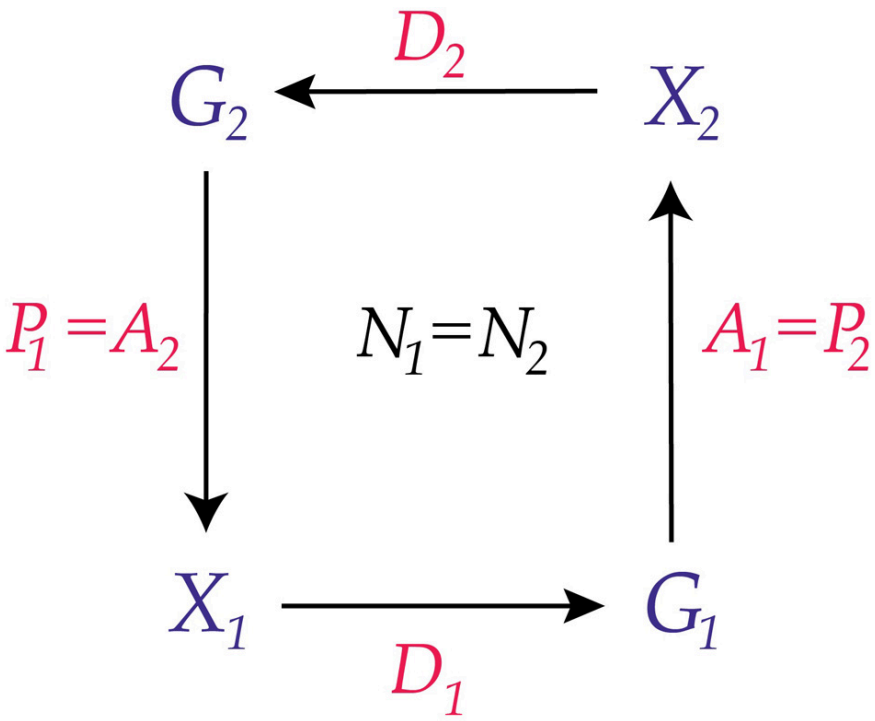
Let

$$C_i = \left(X_i, G_i, P_i, D_i, A_i, N_i\right), \qquad i = 1, 2. \tag{3–4}$$

Suppose their shared world contains only these two agents (Figure 2). Because each belongs to W, it is not strictly an *external* world for either. Connecting them requires

$$P_1 = A_2, \qquad P_2 = A_1, \qquad N_1 = N_2. \tag{5–7}$$

Thus each agent's action channel is the other's perception channel, and both operate synchronously. Agents satisfying Figure 2's commuting diagram are *joined* or *adjacent*: each agent's experiences and actions affect the probabilities of the other's experiences and actions. We represent this reciprocal connection by the undirected edge C_1—C_2. It is an *undirected combination*, developed further in §7.

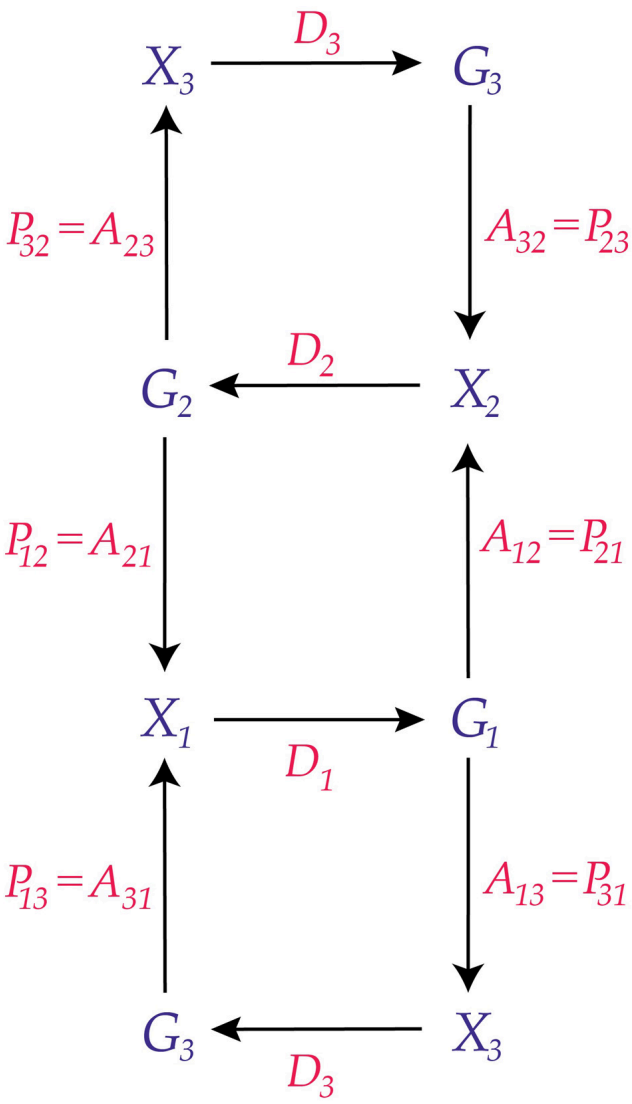


Original Figure 2. Two reciprocally joined agents. Each action channel is the other agent's perception channel.

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Three conscious agents

Add a third agent of the same form, so that the index range becomes $i = 1, 2, 3$, as in equation (8). Figure 4 connects every pair; this graph is *complete*. The two appearances of C_3 in that diagram represent the same agent.



Original Figure 4. Three agents with every pair connected. The two depictions of C_3 represent the same agent.

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To describe C_1's perceptions from C_2 and C_3, let $(X_{12}, \mathcal{X}_{12})$ and $(X_{13}, \mathcal{X}_{13})$ be the respective measurable experience spaces. They need not contain the same kinds of experiences. Define

$$P_1 = P_{12} \otimes P_{13} : (G_2 \times G_3) \times \mathcal{X}_1 \to [0, 1], \tag{9}$$

$$\mathcal{X}_1 = \sigma\big(\mathcal{X}_{12} \times \mathcal{X}_{13}\big), \tag{10}$$

where the latter is the σ-algebra generated by measurable rectangles. On such rectangles the tensor product is

$$P_1\big(\big(g_2, g_3\big), E_{12} \times E_{13}\big) = P_{12}\big(g_2, E_{12}\big) P_{13}\big(g_3, E_{13}\big), \tag{11}$$

with $g_j \in G_j$ and $E_{1j} \in \mathcal{X}_{1j}$, for $j = 2, 3$. This factorization expresses probabilistic independence of the two perceptual outputs for fixed inputs g_2, g_3, while allowing their experience spaces to differ.

Likewise, C_1 may act differently on the two recipients. Let $\big(G_{12}, \mathcal{G}_{12}\big)$ and $\big(G_{13}, \mathcal{G}_{13}\big)$ be its respective measurable action spaces. Define

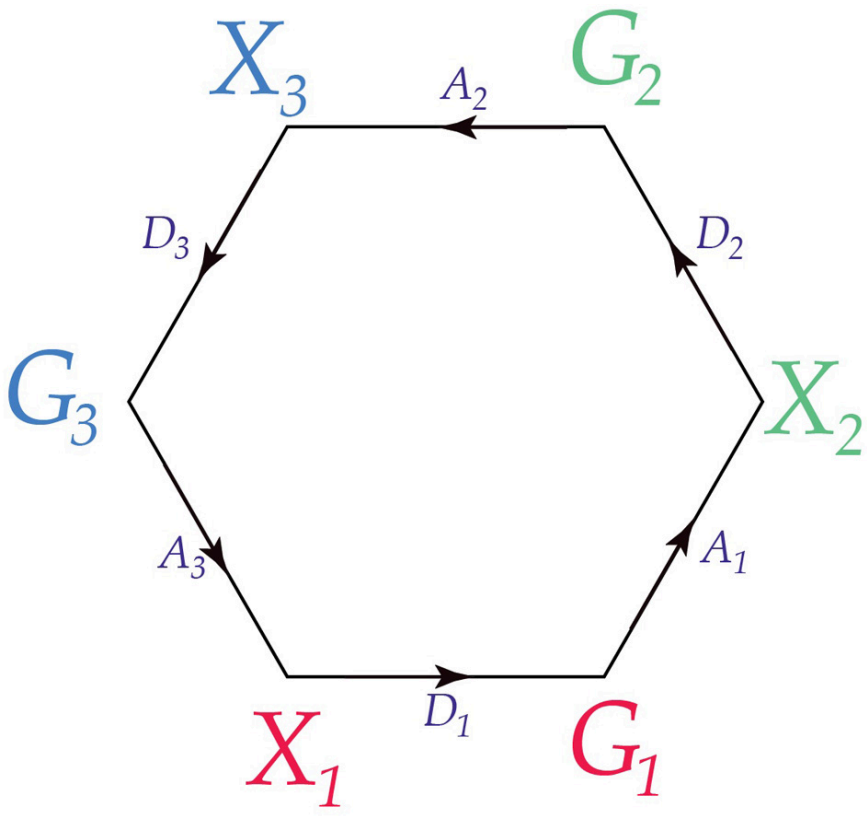
$$A_1 = A_{12} \otimes A_{13} \colon G_1 \times \sigma\big(\mathcal{X}_2 \times \mathcal{X}_3\big) \to [0, 1], \tag{12}$$

$$G_1 = G_{12} \times G_{13}. \tag{13}$$

The preceding joins are undirected: communication occurs in both directions along each edge. *Directed joins* also permit one-way communication. Figure 7 gives the cycle

$$C_1 \to C_2 \to C_3 \to C_1, \qquad A_1 = P_2, \quad A_2 = P_3, \quad A_3 = P_1,$$

with no reverse connections. Each agent sends to its successor and receives from its predecessor.



Original Figure 7. A directed cycle of three agents. Each sends to its successor and receives from its predecessor.

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Directed joins can also represent successive levels of visual processing (Marr, 1982): C_1 solves the correspondence problem to construct two-dimensional motion; C_2 uses those motions to infer three-dimensional structure; C_3 uses that structure for object recognition.

More generally, any number of agents may form any multigraph whose vertices have finite degree: vertices represent agents, and edges represent directed or undirected joins (Chartrand and Ping, 2012). Such networks include deterministic or nondeterministic cellular automata (Ceccherini-Silberstein and Coornaert, 2010) and universal Turing machines (Cook, 2004).

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6. Dynamics of two conscious agents

The two agents defined in (3–4), repeated as (14–15) in the source, form a dynamical system when joined as in Figure 2. Their joint state space is $E = X_1 \times G_1 \times X_2 \times G_2$, equipped with the product σ-algebra $\mathcal{E}$. At step $t \in \mathbb{N}$, the state is $e = \big(x_1(t), g_1(t), x_2(t), g_2(t)\big)$. Each agent simultaneously uses its current perception to choose its next action through D_i, and executes its current action through A_i, affecting the other's next perception.

The transition kernel is

$$L: E \times \mathcal{E} \to [0,1], \tag{16}$$

with, for any measurable next-state event $B \in \mathcal{E}$,

$$\begin{aligned} L(e,B) = \int_B \; & A_2(g_2(t), dx_1(t+1)) D_1(x_1(t), dg_1(t+1)) \\ & \times A_1(g_1(t), dx_2(t+1)) D_2(x_2(t), dg_2(t+1)). \end{aligned} \tag{17}$$

The four kernels multiply to give a joint transition: they act simultaneously and independently, given the current state. Equation (17) does not compose them into successive transitions within a step.

First example: identity kernels

Let each of X_1, G_1, X_2, G_2 contain two states, $|0\rangle$ and $|1\rangle$, and let A_2, D_1, A_1, D_2 all be 2×2 identity matrices. The joint system has $2^4 = 16$ states, written $|x_1 g_1 x_2 g_2\rangle$, from $|0000\rangle$ to $|1111\rangle$.

Its asymptotic, or long-term, behavior is characterized by absorbing sets and their periods. Here each absorbing set is a minimal collection of states that the dynamics cannot leave; its states repeat cyclically. The complete list follows. Within each row, states occur in the order shown, with the last returning to the first.

	Period	Absorbing set, in cycle order
	1	$\|0000\rangle$
	1	$\|1111\rangle$
	2	$\|0101\rangle$, $\|1010\rangle$
	4	$\|0001\rangle$, $\|1000\rangle$, $\|0100\rangle$, $\|0010\rangle$
	4	$\|0011\rangle$, $\|1001\rangle$, $\|1100\rangle$, $\|0110\rangle$
	4	$\|0111\rangle$, $\|1011\rangle$, $\|1101\rangle$, $\|1110\rangle$

Second example: one flip kernel

Keep the other kernels unchanged but replace D_1 with

$$D_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

The system now has two absorbing sets, both of period 8. Each sequence below repeats in the displayed order:

1. $|0000\rangle$, $|0100\rangle$, $|0110\rangle$, $|0111\rangle$, $|1111\rangle$, $|1011\rangle$, $|1001\rangle$, $|1000\rangle$.
2. $|0001\rangle$, $|1100\rangle$, $|0010\rangle$, $|0101\rangle$, $|1110\rangle$, $|0011\rangle$, $|1101\rangle$, $|1010\rangle$.

Replacing D_2, A_1, or A_2 instead gives the same asymptotic cycle structure. Thus an asymptotic behavior can correspond to an equivalence class of interacting agents.

Possible dynamics expand with the richness of E and its kernels, and expand further for three or more agents with different directed and undirected joins, forming connected multigraphs or amenable groups. This motivates the following hypothesis.

Hypothesis 2 — Conscious-agent thesis. Every property of consciousness can be represented by a property of a dynamical system of conscious agents.

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7. The combination problem

Any theory taking consciousness as fundamental must explain how elementary consciousness combines into the complex consciousness we possess (Seager, 1995; Goff, 2009; Blamauer, 2011; Coleman, 2014). James's objection is that assembling 100 private feelings leaves 100 separate feelings. A 101st feeling belonging to the group would be a new fact, neither identical with nor deducible from them. Their assembly might trigger its creation under a law, without explaining how they constitute it (James, 1890/2007).

There are two problems. **Combining experiences** concerns how qualia produce a new experience within one subject: how, for example, tastes of salt, garlic, onion, basil, and tomato become the taste of pasta sauce. **Combining subjects** concerns how distinct points of view form a new, unified point of view. Existing work offers no rigorous solution to either. Coleman (2014) considers qualitative combination promising, provided qualities blend or pool: merely preserving each elementary quality

would not explain the homogeneity of experiences such as color. Subject combination faces stronger objections. Nagel (1979) suggests that a point of view's components need not themselves possess points of view; Coleman argues that micro-subjects cannot constitute a macro-subject.

Conscious-agent theory offers two constructive forms of combination. For both proofs, write the constituent agents and proposed composite as

$$\begin{aligned} C_i &= \big((X_i, \mathcal{X}_i), (G_i, \mathcal{G}_i), P_i, D_i, A_i, N_i\big), \quad i = 1, 2, \\ C &= ((X, \mathcal{X}), (G, \mathcal{G}), P, D, A, N). \end{aligned} \tag{18–20; 30–32}$$

Theorem 1: Undirected Join Theorem

An undirected join of two conscious agents creates a new conscious agent.

Proof by construction. Suppose C_1 and C_2 have an undirected join. Set

$$\begin{aligned} X &= X_1 \times X_2, & G &= G_1 \times G_2, \\ P &= P_1 \otimes P_2 \colon G^T \times \mathcal{X} \to [0,1], \\ D &= D_1 \otimes D_2 \colon X \times \mathcal{G} \to [0,1], \\ A &= A_1 \otimes A_2 \colon G \times \mathcal{X}^T \to [0,1], \\ N &= N_1 = N_2. \end{aligned} \tag{21–26}$$

Here T reverses the order of the factors, so $X^T = X_2 \times X_1$ and $G^T = G_2 \times G_1$. The event spaces are the product σ-algebras $\mathcal{X} = \sigma\{E_1 \times E_2 \colon E_i \in \mathcal{X}_i\}$ and $\mathcal{G} = \sigma\{B_1 \times B_2 \colon B_i \in \mathcal{G}_i\}$. In the discrete case, the product kernels are explicitly

$$\begin{aligned} P\Big(\big(g_2, g_1\big), \big(x_1, x_2\big)\Big) &= P_1\big(g_2, x_1\big) P_2\big(g_1, x_2\big), \\ D\Big(\big(x_1, x_2\big), \big(g_1, g_2\big)\Big) &= D_1\big(x_1, g_1\big) D_2\big(x_2, g_2\big), \\ A\Big(\big(g_1, g_2\big), \big(x_2, x_1\big)\Big) &= A_1\big(g_1, x_2\big) A_2\big(g_2, x_1\big), \end{aligned} \tag{27–29}$$

where $g_i \in G_i$ and $x_i \in X_i$. With these choices, C satisfies the definition of a conscious agent. □

This agent is the constituents' **undirected combination** (Figure 2). The construction extends to more than two agents with an undirected join, yielding one composite agent.

Theorem 2: Directed Join Theorem

A directed join of two conscious agents creates a new conscious agent.

Proof by construction. Suppose $C_1 \to C_2$. Using the same six-tuple notation, set

$$\begin{aligned} X &= X_1, & G &= G_2, \\ P &= P_1, & A &= A_2, \\ D &= D_1 A_1 D_2 : X_1 \times \mathcal{G}_2 \to [0,1], \\ N &= N_1 = N_2. \end{aligned} \tag{33–38}$$

The juxtaposition $D_1 A_1 D_2$ denotes kernel composition. In standard measure notation, Eq. (36) is

$$D(x_1, B) = \int_{G_1} \int_{X_2} D_2(x_2, B)\, A_1(g_1, \mathrm{d}x_2)\, D_1(x_1, \mathrm{d}g_1), \qquad B \in \mathcal{G}_2.$$

With these choices, C satisfies the definition of a conscious agent. □

This is the agents' **directed combination**. The construction extends to several agents having directed joins to C_2: all of them, together with C_2, form a new agent.

Conjecture 3: Combination Conjecture. In any pseudograph of conscious agents, with any mixture of directed and undirected edges, any subset of agents—whether adjacent or not—can be combined into a new conscious agent.

Combining experiences

An undirected combination has experiential space $X_1 \times X_2$, allowing experiences unavailable to either constituent. If C_1 experiences only achromatic brightness and C_2 only hue variations, their combination experiences hues of varying brightness. The Cartesian-product structure does not preclude perceptual dependence between the components: feedback in the undirected join can generate it (Maddox and Ashby, 1996; Ashby, 2000).

A directed combination retains $X = X_1$, but X_2 still contributes through the decision kernel $D_1 A_1 D_2$, which literally integrates over that entire experiential space. Evidence that decision-making has a phenomenology (Nahmias et al., 2004; Bayne and Levy, 2006) suggests that what deciding feels like may be intimately related to the experiential spaces integrated into a decision. The formalism thus suggests a prediction and a requirement for solving the experiential combination problem: perceptual experience must be considered together with decision-making.

Combining subjects

For Coleman (2014), a subject is a point of view with a private qualitative field, to which other subjects have no direct experiential access. A conscious agent C_i meets this description: it has its own experiential sphere X_i, lacks direct experiential access to another agent's X_j, and additionally makes decisions and takes actions. The theory therefore identifies a subject or point of view with a six-tuple satisfying the conscious-agent definition.

Goff's objection is that the existence of subjects with specified phenomenal characters never *a priori* entails another subject (2009). Coleman argues that combination is positively incoherent: each point of view is essentially unified and experientially isolated, so a collection is essentially diverse; extracting one unified successor from that diversity would require emergence, with the original viewpoints contributing nothing as viewpoints (2014).

On the theory's identification of subjects with agents, the join theorems construct new points of view from existing ones. The constituents retain their identities while influencing one another; their structures and interactions make mathematically intelligible contributions to the composite's properties. New properties need not be brute emergence. An undirected combination, for instance, can exhibit periodic asymptotic behavior absent from either constituent but explicable from their interaction.

The authors present this as the first rigorous theoretical account of subject combination, whose further deductive implications remain to be explored. One suggested implication is **introspection**: the undirected combination is an agent whose world W is itself, suggesting how introspection could arise intelligibly through combination.

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8. Microphysical objects

Space-time and objects can be understood as symbols through which conscious agents represent one another's properties and interactions. Assuming each agent is finite and the realm of interacting agents infinite, an agent's representation must simplify; it therefore functions as an interface rather than an isomorphism. We propose that microphysical objects represent the asymptotic, or long-term, dynamics of interacting agents, with space-time furnishing a representational framework. The proposed connection is that harmonic functions associated with these dynamics have the form of a free-particle wavefunction: particles would represent vibrations of interacting conscious agents.

The two-agent dynamics governed by the kernel L in (17) is Markovian: transitions depend only on the current state. Its associated *space-time chain* has kernel

$$Q\colon (E\times\mathbb{N})\times\left(\mathcal{E}\otimes 2^{\mathbb{N}}\right)\to[0,1], \tag{39}$$

$$Q((e,n),A\times\{m\})=\begin{cases} L(e,A), & m=n+1,\\ 0, & \text{otherwise,}\end{cases} \tag{40}$$

where $e\in E$, $n,m\in\mathbb{N}$, and $A\in\mathcal{E}$. The quasi-compactness theorem invoked here (Revuz, 1984), which the paper takes to apply by finiteness of the state space, gives a cyclic asymptotic decomposition:

1. Finitely many invariant, absorbing events, indexed by ρ, exhaust E. Once entered, an absorbing event cannot be left.
2. Each event ρ partitions into d_ρ asymptotic events. Their indices advance with certainty through $\delta=1,\ldots,d_\rho$, cyclically.

Let $U_{\rho,\delta}$ indicate the corresponding asymptotic event, and let k be an integer modulo d_ρ. Define

$$\lambda_{\rho,k}=\exp\left(2i\pi k/d_\rho\right), \tag{41}$$

$$f_{\rho,k}=\sum_{\delta=1}^{d_\rho}\left(\lambda_{\rho,k}\right)^{\delta}U_{\rho,\delta}. \tag{42}$$

In the first asymptotic example of Section 6, the six absorbing sets have periods $\left(d_1,\ldots,d_6\right)=(1,1,2,4,4,4)$. Here $U_{1,1}$ indicates $|0000\rangle$, and $U_{5,3}$ indicates $|1100\rangle$: each equals one on its specified state and zero elsewhere.

The eigenfunction–harmonic-function correspondence (Revuz, 1984, p. 210) gives

$$Lf_{\rho,k}=\lambda_{\rho,k}f_{\rho,k}, \tag{43}$$

and the Q-harmonic function

$$g_{\rho,k}(\,\cdot\,,n)=\left(\lambda_{\rho,k}\right)^{-n}f_{\rho,k}. \tag{44}$$

Substitution of (41)–(42) yields

$$
\begin{aligned}
g_{\rho,k}(\cdot, n) &= \exp\left(2i\pi k/d_\rho\right)^{-n} \sum_{\delta=1}^{d_\rho} \exp\left(2i\pi k/d_\rho\right)^{\delta} U_{\rho,\delta} \\
&= \sum_{\delta=1}^{d_\rho} \exp\left(2i\pi k\frac{\delta}{d_\rho} - 2i\pi k\frac{n}{d_\rho}\right) U_{\rho,\delta} \\
&= \sum_{\delta=1}^{d_\rho} \operatorname{cis}\left(2\pi\frac{k\delta}{d_\rho} - 2\pi\frac{kn}{d_\rho}\right) U_{\rho,\delta} \\
&= \sum_{\delta=1}^{d_\rho} \operatorname{cis}\left(2\pi\frac{\delta}{d_{\rho,k}} - 2\pi\frac{n}{d_{\rho,k}}\right) U_{\rho,\delta},
\end{aligned}
\tag{45}
$$

where $d_{\rho,k} = d_\rho/k$, and $\operatorname{cis}\theta = \cos\theta + i\sin\theta$. This has the form of the free-particle wavefunction presented by Allday (2009, §7.2.3):

$$
\psi(x,t) = A\sum_x \operatorname{cis}\left(2\pi\frac{x}{\lambda} - 2\pi\frac{t}{T}\right)|x\rangle. \tag{46}
$$

We therefore propose the following identifications:

Particle description	Agent dynamics
Wavefunction ψ	Harmonic function $g_{\rho,k}$
Amplitude A	1
Position basis $\|x\rangle$	Event indicators $U_{\rho,\delta}$
Position index x	Asymptotic-event index δ
Time t	Step number n
Wavelength λ and period T	$d_{\rho,k}$

The resulting assignments are $p = h/d_{\rho,k}$ and $E = hc/d_{\rho,k}$, where h is Planck's constant and c the speed of light. Both momentum and energy are inversely proportional to $d_{\rho,k}$. Since $\lambda = T$, the wave has unit speed in the chosen units. The paper describes this as a nonrelativistic identification.

For a relativistic extension, consider an undirected join with six state components: $N_1, N_2, X_1, X_2, G_1, G_2$. We propose identifying these with generating vectors of the geometric algebra $\mathcal{G}(2,4)$, assigning positive signature to N_1, N_2 and negative signature to the other four. This is the conformal geometric algebra associated with Minkowski space of signature $(1,3)$ (Doran and

Lasenby, 2003). Its conformal group contains the Poincaré group of space-time translations and rotations; the full conformal group is needed for most massless relativistic theories and appears in supersymmetry and supergravity. The rotor group's isomorphism with $SU(2,2)$ connects this proposal with Penrose's twistor program for quantum gravity (Penrose, 2004).

Each undirected join would supply a new $\mathcal{G}(2,4)$ and new generating vectors. The proposed nested hierarchy would construct space-time and massless particles from Planck to macroscopic scales, with the metric arising from the joined agents' channel capacities.

For massive particles, which require symmetry breaking in this proposal, a possible route is a hierarchy of stopping times. Infrequent interactions between systems can be modeled by stopping times and, through the combination theorems, create agents whose clocks advance more slowly than those of their constituents. The proposed hierarchy of slowing clocks accompanies a hierarchy of agent combinations, extending to human conscious experience—roughly 10^{40} times slower than the Planck time—and beyond.

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9. Objections and replies

1. Does the definition distinguish consciousness?

Objection. The definition could describe unconscious agents equally well, so it says nothing about consciousness.

Reply. Applicability to unconscious systems would not prevent the same mathematical structure from modeling conscious systems. A formalism can serve several domains; using integers to count one kind of thing does not prevent counting another. Objection 21 develops this defense.

2. What does a mathematical model leave out?

Objection. Formalizing consciousness must lose something essential.

Reply. It does leave out consciousness itself, just as a mathematical hurricane produces no rain. A model need not instantiate its subject to reveal that subject's structure. The proposed mathematics represents consciousness; it is not identical to consciousness.

3. Why represent qualia probabilistically?

Objection. Why use a probability space X for conscious experiences?

Reply. Probability theory's application to other domains does not disqualify its application to qualia. The representation assumes that experiences convey information, drawing on Shannon's connection between information and probability. It imposes no universal requirement that experiences possess a metric or dimensions, although particular experience spaces, such as phenomenal color, may possess both.

An agent C's experiences X are private: another agent C_i experiences only its own X_i, never X directly. In this sense, X is internal to C; its exterior is W, more precisely $W \backslash C$.

4. Where is free will?

Objection. A conscious agent should have free will.

Reply. For an experience $x \in X$, the decision kernel $D(x, \cdot)$ assigns probabilities to available actions in G. These probabilities are interpreted as the agent's free choices and as objective nondeterminism in nature. The authors tentatively interpret the remaining probabilities subjectively, as ignorance or degrees of belief.

5. Where are goals?

Objection. The definition should include goals and purposeful behavior.

Reply. Goals are omitted because consciousness need not, by definition, be goal-directed. Nevertheless, systems of conscious agents can implement universal Turing machines and therefore can implement goal-directed behavior. When a goal is consciously experienced as a desire, it can be represented in X.

6. Are permanent conscious agents still permanent objects?

Objection. Conscious agents continue existing unobserved and generate perceptions of shape and size, so the theory retains object permanence.

Reply. Conscious realism accepts agents existing independently of a particular observer, thereby rejecting solipsism. It rejects *physical* object permanence: unperceived three-dimensional space and physical objects. The continued existence of conscious agents is a different claim from the continued existence of unconscious objects and spacetime.

7. Can science work through an interface?

Objection. If perceptions do not resemble objective reality, scientific investigation becomes impossible.

Reply. Constructing theories and testing predictions can continue. Perceptual resemblance is one theory that can be rejected and replaced; it is not a prerequisite of scientific method. Evidence already underdetermines theory: no body of data uniquely identifies the correct explanation. Interface theory introduces no special version of that familiar problem.

8. Does fundamental consciousness fall outside science?

Objection. Making consciousness more fundamental than physics places it beyond science.

Reply. The proposal incurs three scientific obligations: formulate consciousness rigorously, recover current physics as a special case, and predict testable results beyond current physics. Rejecting fundamental spacetime and objects rejects a particular physicalist theory, not those methodological obligations.

9. Does the evolutionary argument undermine its own reasoning?

Objection. Following Plantinga (2002), unreliable perception undermines confidence in cognition and therefore in natural selection itself.

Reply. Results about selection on perception do not establish the reliability of every cognitive faculty. Selection might favor accurate logic and mathematics because estimating the fitness consequences of actions requires them. Reliability must therefore be investigated faculty by faculty; the perceptual simulations cannot settle the general question.

10. Does a product of decision kernels combine subjects?

Objection. An undirected join merely multiplies the agents' decision kernels, representing two independent decisions rather than one decision-making subject.

Reply. The proposed combination is a process. Decisions initially factorize, but continuing feedback couples the agents increasingly. The authors claim that the asymptotic decision structure generally ceases to factor into the original kernels, giving the combined agent an integrated decision structure. Its constituents retain their identities while participating in this new agent. Objection 18 supplies the dynamical examples and proposed quantification supporting this response.

11. Why regard consciousness itself as real?

Objection. If neurons are perceptual icons, perhaps consciousness is another icon rather than an absolute reality.

Reply. Conscious realism explicitly assumes that W consists of conscious agents, and the agent formalism assumes that qualia exist. These are commitments of the theory. They do not entail either *incorrigibility*—believing one has an experience guarantees having it—or infallibility about experience's contents. The authors cite change blindness as evidence against those stronger claims. Nor do they identify their mathematics with consciousness.

One could investigate a psychophysical-monist interpretation in which matter and consciousness are aspects of a more abstract reality. Even if possible, it would retain the claim that dynamical physical properties such as position, momentum, and spin have definite values only when observed, making it objectionable to many physicalists.

12. What exactly are perception and truth?

Objection. Neither term is defined here. A cautious predator-avoidance system may produce false alarms, but surely encountering a real predator still brings it into contact with truth.

Reply. The formal definitions appear in the cited simulation studies. A perceptual strategy is a measurable map, or more generally a Markov kernel,

$$p: W \to X,$$

where W represents the objective world and X the organism's possible perceptions. Four nested strategy classes distinguish different kinds of truth:

Strategy	Formal condition	Relationship to truth
Naïve realist	$X = W$; p is an isomorphism preserving every structure on W	Perceives all and only the truth.
Strong critical realist	$X \subseteq W$; p preserves the relevant structure on that subset	Perceives some, possibly not all, of the truth.
Weak critical realist	X need not lie within W; p preserves structure	Perceptual contents need not be world contents, but their relationships reflect objective relationships.

Strategy	Formal condition	Relationship to truth
Interface	Neither inclusion nor structure preservation is required	Contents and relationships need not resemble those of W.

Thus naïve realism is contained in strong critical realism, then weak critical realism, then the interface class. The last class permits nonveridical strategies; its definition does not exclude veridical special cases.

In the reported simulations across hundreds of thousands of randomly selected worlds, truth-oriented strategies tend to lose against fitness-oriented interface strategies; in genetic algorithms they often fail to evolve at all. Perceiving a lion is a real experience, just as a headache is real. It does not establish an observer-independent lion resembling that experience. The theory treats the lion as an adaptive symbol triggered by something in W, without requiring resemblance between symbol and trigger.

13. Does fitness exclude truth?

Objection. If the argument assumes fitness and truth are incompatible, that premise is implausible. If it assumes only that they need not coincide, the conclusion that perception never tracks truth does not follow. Escaping a tiger or caring for children seems to require truth; otherwise why have perception rather than arbitrary kaleidoscopic experience?

Reply. Neither formulation is the intended argument. The starting point is the simulation result in Objection 12. Fitness is represented as

$$f: W \times O \times S \times A \to \mathbb{R},$$

depending on the world, organism, organism's state, and action. Because fitness is not a function of world structure alone, it must be distinguished from truth. The authors maintain that truth-oriented strategies survive when fitness happens to vary monotonically with relevant world structure; otherwise selection favors fitness-oriented strategies.

Their proposed argument is: selection favors fitness; fitness *generically* diverges from truth; therefore selection generically favors perceptions that diverge from truth. Here "generic" means outside a set of measure zero, relative to an appropriate unbiased measure. A line has zero area within a plane despite containing infinitely many points; likewise, exceptional cases need not be nonexistent.

The conclusion is therefore not that selection *never* favors truth. Some human perceptions could be exceptional, although the authors regard this as unlikely. Children and predators receive the same treatment as the lion in Objection 12: their perceptual forms guide consequential behavior without establishing matching public physical objects. Perception avoids arbitrariness because it tracks fitness.

14. Is there a liar paradox?

Objection. A perception-based claim that perception never concerns truth seems to undermine its own truth, producing a contradiction.

Reply. Let E be the hypothesis that perception evolved through natural selection. If E holds, the authors' evolutionary argument predicts generically fitness-oriented perception. They present this as a testable hypothesis, with further proposed consequences: physical objects lack intrinsic causal powers and have no definite dynamical properties while unobserved. They identify these consequences with features of standard quantum interpretations.

If E fails, the paper says science then has no alternative account of perceptual development from which to assess veridicality. Neither branch creates a logical contradiction. Science must be allowed to investigate human perception's accuracy just as it investigates other species; Marr's account of a fly's subjective, action-centered visual representation illustrates that existing practice.

15. Doesn't scientific correction reveal more truth?

Objection. Replacing a flat Earth with a round Earth shows that some perceptions prevail because they are true.

Reply. The authors interpret successive models as useful across wider contexts. A flat approximation serves many everyday tasks, a sphere supports circumnavigation, and satellite control or submarine navigation demands greater complexity. That progression does not, on their view, establish resemblance to reality.

Their evolutionary argument, alongside related embodied-cognition research, treats perceived space as an interface. Physics extends this space through mathematical transformations, including Euclidean and Poincaré groups and general differentiable coordinate transformations. Spatial objects and their shapes consequently remain descriptions within the species' representational vocabulary.

16. Can icons be taken literally as icons?

Objection. Icons really are icons; there is nothing wrong with taking that fact literally.

Reply. Agreed. A perceived red apple exists as a perceptual icon, which disappears when the observer closes their eyes. The disputed inference is that a matching objective red apple continues existing unperceived and produced the icon. The evolutionary argument is directed at that inference.

17. Why reject physical causation?

Objection. Causal belief is deeply ingrained, and even if causation disappears microscopically, macroscopic causation may remain. Appealing to Hume does not establish otherwise.

Reply. A psychological disposition toward causal belief, whether innate or acquired early, does not establish its accuracy. The authors acknowledge sharing it, but regard their evolutionary results as grounds for revising both physical realism and apparent causation between physical objects.

They allow that perceptual icons inform behavior and might, in that sense, have causal powers. This differs from the causation ordinarily attributed to independently existing physical objects. Their position derives principally from evolutionary modeling, not from Hume, whose interpretation remains disputed.

They regard their account as compatible with Copenhagen, quantum Bayesian, and arguably many-worlds interpretations that they understand as abandoning microphysical causality. They place the explanatory burden on proposals retaining macroscopic causation despite abandoning its microphysical counterpart.

18. Is the proposed combination merely asserted?

Objection. Cartesian products do not produce genuine experiential unity. Objection 10's claims about growing dependence and an unspecified limit merely assert the required result.

Reply. The intended limit is the technical asymptotic regime of Markov-chain theory. The two examples in "Dynamics of two conscious agents" exhibit undirected joins whose long-run behavior does not reduce to a simple product of constituent behavior. Feedback allows each agent's decisions and actions to affect the other. Across iterations, the joint dynamics organize into absorbing sets with characteristic periodic behavior.

The authors propose quantifying departure from product behavior using integrated information and report two bits for the second example: information attributable to the combined dynamics beyond that of the isolated agents.

Using this measure does not commit them to Tononi's identification of consciousness with integrated information or of qualia with informational relationships. They also reject its consequence that spectrum inversion is impossible. They invoke Hoffman (2006) as proving reductive functionalist

theories false and therefore regard this identification as untenable. A weaker proposal that integrated information *causes* consciousness would escape that particular objection, but would require an explanatory theory connecting the two. The authors say none is supplied and doubt that such a theory is possible.

19. Does denying public objects undermine evolution and the models?

Objection. The paper privileges its own account while supposedly denying reality. If organisms, genes, and computers are merely perceptual, its evolutionary reasoning and simulations seem equally disqualified.

Reply. The paper accepts objective reality and asks how human perception relates to it. It uses established science to infer that reality probably does not share our perceptions' structure, then proposes interacting conscious agents as an alternative ontology.

That proposal requires revisiting evolution. Standard evolutionary theory describes organisms and genes in spacetime, which conscious realism treats as a perceptual representation. A generalized theory would instead describe evolving systems of conscious agents. Its crucial constraint is that projection into the human spacetime interface must recover standard evolutionary theory. The latter becomes a boundary condition and successful representation, without being assumed literally fundamental.

This is the authors' instrumentalist account of scientific progress: an effective current theory can support construction of a successor that explains both its successes and its limitations. The successor must explain why the earlier evolutionary framework enabled the transition.

Their proposed research program retains universal Darwinism's abstract variation, selection, and retention, applying them to agent networks. Information, quantified through channel capacity, would be the limited resource driving competition. They motivate this through a claimed information–energy equivalence and information-to-energy conversion, paralleling organisms' competition for energy such as food. This generalized evolutionary theory is a project, not a completed result here.

20. Is truth versus reproductive success a false dichotomy?

Objection. Contrasting truth with fitness or having children imposes a false dichotomy.

Reply. As §§3 and 9.13 formalize, one objective world supports different fitness functions for different organisms, states, and actions. Those functions select different perceptual strategies; truth alone does not specify which strategy benefits an organism.

21. Doesn't mathematical generality still leave consciousness unexplained?

Objection. Integers can count apples, oranges, and treaties precisely because they are not theories of those things. Likewise, a definition applying equally to conscious and unconscious agents may define only agency. A definition broad enough to cover complex numbers would not uniquely define integers.

Reply. A model's adequacy depends on appropriate structure and successful empirical predictions, not exclusive applicability. Integers lack the structure needed for a theory of treaties; their applicability to counting other things is not the problem.

The same mathematical structure can model distinct subjects: $SU(3)$ describes exact quark-color and approximate flavor symmetries; a Markov kernel can describe both stochastic dynamics and a communication channel. Similarly, X might represent a human's conscious color experiences and a robot's unconscious color judgments. Successful application to the latter does not logically exclude the former. The conscious-agent formalism should therefore be judged by its fit to consciousness and its predictions, rather than rejected merely because it can represent unconscious systems.

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10. Conclusion

Object permanence encourages explaining perceiver-independent objects before subjects. Yet the reported evolutionary games and genetic algorithms generally favor fitness-oriented over veridical perception, suggesting that perceived objects are icons of a species-specific interface. The resulting framework cannot assume that physical objects possess genuine causal powers or that space-time is fundamental.

If conscious subjects are fundamental, a mathematically precise account must recover objects and all physics from them. This paper takes three initial steps: (1) formalizing conscious agents; (2) using that formalism to find solutions to the combination problem; and (3) sketching how agents' asymptotic dynamics might generate particles and space-time. Completing this account remains a substantial task. Its success would replace object permanence as our starting point with objects of consciousness.

Editorial notes

These notes distinguish minor corrections from unresolved points in the original. They are additions to the condensed adaptation.

1. **Notation corrected.** The channel probability is $p(y \mid x)$, the probability of output y given input x; §4 prints the conditional in reverse. Equation (28) requires $D_2(x_2, g_2)$, not the printed $D_2(g_1, g_2)$. An extra parenthesis in the six-tuple and the inconsistent time label for newly chosen actions are corrected. Calligraphic letters consistently distinguish event σ-algebras from state sets. The normalized fitness measure, the two defining properties of a Markov kernel, and the integral for composed kernels are explicit restatements of the mathematical constructions.

2. **Join constructions retain an interface issue.** Definition 1 requires a single world space W for the input of P and output of A. In the undirected construction, the displayed P takes input in G^T, whereas A outputs to X^T. The source does not explicitly supply the identifications or common world needed to reconcile these types in full generality. The directed construction likewise needs compatible interfaces. The theorem statements and constructions are retained as the paper presents them; their interpretation as combination of conscious subjects additionally uses its identification of subjects with these formal agents.

3. **Feedback and limits.** Objection 10 writes D^n, although $D: X \times \mathcal{G} \to [0, 1]$ need not be composable with itself when $X \neq G$. Its intended claim is retained as a claim about the long-run joint dynamics. Periodic Markov dynamics need not approach a single pointwise limit, and feedback alone does not establish monotonic growth of dependence. The paper reports two bits of integrated information in Objection 18 without presenting the calculation there.

4. **Scope of the cyclic decomposition.** Even for finite E, the space-time chain Q has the infinite state space $E \times \mathbb{N}$. Finiteness of E therefore does not itself establish the paper's stated quasi-compactness justification for Q. Closed recurrent classes also need not cover transient states of a general finite chain, although the permutation examples in §6 have no transient states. The decomposition is retained in the form invoked by the source.

5. **The particle correspondence remains a proposal.** The expression $d_{\rho,k} = d_\rho / k$ is undefined at $k = 0$; the earlier eigenfunction formula still handles that mode. For integer k, the discrete phase's fundamental period is $d_\rho / \gcd(k, d_\rho)$, which need not equal d_ρ / k. The source's assignments $p = h / d_{\rho,k}$ and $E = hc / d_{\rho,k}$ imply $E = cp$, rather than the usual massive nonrelativistic dispersion $E = p^2 / (2m)$. Its matching of harmonic forms and proposed identifications therefore does not by itself derive that dynamics. The geometric-algebra, metric, mass, and generalized-evolution constructions remain proposed extensions.

6. **Qualification retained across sections.** The more categorical evolutionary wording in the opening sections is read together with §9, objection 13: the authors allow exceptional cases in which selection favors truthful perception. Their measure-theoretic use of “generic” depends on an appropriate measure, which is not constructed in this paper. Claims about quantum interpretations, reductive functionalism, and information–energy equivalence remain attributed to the paper; this condensation supplies no new evidence for them.

Original references

The source bibliography is retained for tracing the paper’s arguments. Inclusion here does not imply that each cited work was independently reviewed for this edition.

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Source text and original figures © 2014 Hoffman and Prakash, used and adapted under Creative Commons Attribution 3.0. Changes: prose condensation, consolidation of repeated definitions and diagrams, standardized mathematical notation, and separate editorial notes.
Word comparison covers prose in the abstract and ten main sections; displayed equations, section headings, source figure captions, the bibliography, conflict statement, and editorial notes are excluded. Inline mathematical notation is omitted from the condensed count. Figures 1, 2, 4, and 7 retain the distinct source diagrams; alternative drawings of the same connections are omitted.